Lead-Lag Relationships Between Regional Oil Prices, US Retail Oil Prices, and Crude Oil Prices

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Executive summary

This report analyzes the relationship of American gasoline prices with crude oil prices and the comparison of regional gasoline prices within the United States in order to model, forecast and understand gasoline price dynamics in the United States. These findings are based on an initial ARMA (auto-regressive and moving average) fit on the gasoline prices and two fitted VAR (vector autoregressive) models. The first VAR model is fit on US gasoline prices and Brent crude oil prices. The second VAR model is fit on the gasoline prices for 5 subregions of the United States: West Coast, Rocky Mountains, Midwest, Gulf Coast, and East Coast. Based on the findings, the VAR (vector autoregressive) model should that gasoline prices does have a dependence on the one-week lag from oil prices but the predictions from the ARMA (auto-regressive and moving average) are more accurate. Changes in the price of Brent crude oil from a week ago affect the price of gasoline in the US. Regional gasoline prices overall tend to move in conjunction with each other, but gasoline prices in the West Coast and the Rockies regions seem to be more driven by local impacts to the price, possibly due to higher transportation-related costs and policy-driven costs. Both VAR and ARIMA models predict a decrease in the next four weeks, but ARIMA seems to provide better predictions. The strongest errors in the models are due to external shocks that are outside the scope of this model and could not be forecasted.

1 Introduction

This analysis explores gasoline price dynamics by analyzing the relationship of American gasoline prices with crude oil prices and the comparison of regional gasoline prices within the United States. The main research questions are:

- How do Brent crude oil price changes pass through to U.S. retail gasoline prices, and over what time range does this occur?
- How do regional gasoline prices co-move over time? Are there lead–lag relationships between regions?

This analysis helps explain how crude oil shocks affect consumer fuel prices and inflation. Forecasting this relationship can improve pricing expectations, support policy and business planning, and help anticipate the impact of supply disruptions or geopolitical events.

Based on the findings, the VAR (vector autoregressive) model should that gasoline prices does have a dependence on the one-week lag from oil prices but the predictions from the ARMA (auto-regressive and moving average) are more accurate. Changes in the price of Brent crude oil affect the price of gasoline in the US at a one-week lag. Regional gasoline prices are correlated and move in conjunction with each other, especially in the Midwest, Gulf Coast and East Coast. Gasoline prices in the West Coast and the Rockies regions have less correlation with the other regions, especially during larger shocks, implying more local

impacts to the price. The West Coast and Rockies regions may have higher transportation-related costs and policy-driven costs, such as tax. Forecasts predict a decrease in gasoline and Brent crude oil prices in the short term with the best prediction being the mean in the long term.

1.1 Data

The main data sources are the weekly U.S. gasoline prices from the U.S. Energy Information Administration (U.S. Energy Information Administration 2026) and Brent crude oil prices from Yahoo Finance (2026), which is pulled through Python's yfinance package (Aroussi 2026). The weekly U.S. gasoline prices include gasoline prices for overall country and regional prices. The time period for the data is from 1994 to 2026 at a weekly frequency. This is a total of 1722 observations for each time series. Brent crude is the main input cost for gasoline, so changes in oil prices often pass through to pump prices with a lag. Regional gasoline data also captures differences in refining capacity, transportation, and local supply constraints. The regions are East Coast, Midwest, Gulf Coast, Rocky Mountains, and West Coast.

2 Methods

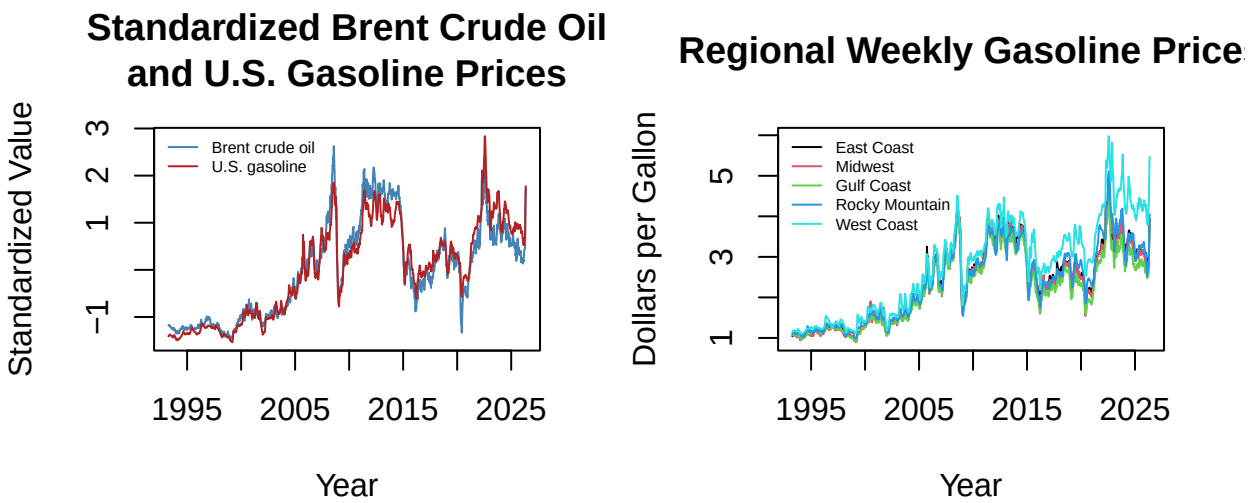

Figure 1: Regional gasoline prices, Brent crude oil prices and U.S. gasoline prices follow a shared general trend with similar rises and drops. There are still discrepancies between different time series.

As shown in Figure 1, regional gasoline prices, Brent crude oil prices and U.S. gasoline prices have similar shocks and directional trend, but there are still discrepancies between different time series. In particular, there are lags between the gasoline and crude oil prices, and the regional prices have varying magnitude for shocks, implying local dynamics influencing the regions individually. Some of the shocks can be explained by the historical events at the time.

The clear trends suggest that the series are non-stationary. Therefore, all time series (i.e. Brent crude oil prices, U.S. gasoline prices, and the five regional gasoline prices) are log-transformed and differenced in order to achieve stationarity (e.g., Figure 2 shows that the transformed U.S. gasoline prices looks approximately stationary).

Our first objective was to fit an ARIMA model to the U.S. gasoline price series in order to understand the structure of the data and create a baseline model. This is achieved by examining ACF and PACF plots and using the `auto.arima` function to identify potential models.

Building on this, we extended the analysis to VAR models in order to incorporate the effect of crude oil prices on gasoline prices. We first examined CCF (cross-correlation function) plots in order to identify the lags with the strongest relationships. The `VARselect` function was used to identify potential models. The same VAR methodology was used to examine the relationships among the regional gasoline prices.

Therefore, this analysis fit two VAR models: one using US gasoline prices and Brent crude oil prices and one using the 5 regional oil prices. These two models are used to answer the two research questions above, respectively. VARselect suggested different ordered models based on several selection criteria, including AIC (Akaike Information Criterion), HQ (Hannan-Quinn Criterion), SC (Schwarz Criterion), and FPE (Final Prediction Error). In both analyses, a tradeoff between the simplest order due to parsimony and model diagnostics was used to choose the final model. The cross-lag coefficients of the VAR models were interpreted in order to identify which series lead and which follow. Diagnostics indicated that the VAR model was appropriate for these time series. Diagnostics include comparing the observed and fitted values (both in relation to time and their correlation to each other) and assessing the normality and stationarity of residuals using QQ-plots and ACF/PACF respectively. We also use residual CCF to assess the VAR models. These models also are used to forecast each time series over a four-week period.

3 Results

3.1 Gasoline Prices ARIMA(4,0,1) Model

Figure 2 shows that the logging and differencing removed trends from the series, which now looks fairly stationary. From Figure 3, we see that the ACF gradually decreased, but rather than declining smoothly it displayed slight oscillation and several small but significant spikes at later lags. The PACF had a strong spike at lag 1, with smaller spikes at the next few lags before cutting off. This pattern suggested that the series may contain both autoregressive and moving average components rather than a simple autoregressive structure alone. There was no clear evidence of seasonality. The `auto.arima` function selected an ARIMA(4,0,1) model with no intercept for the differenced series.

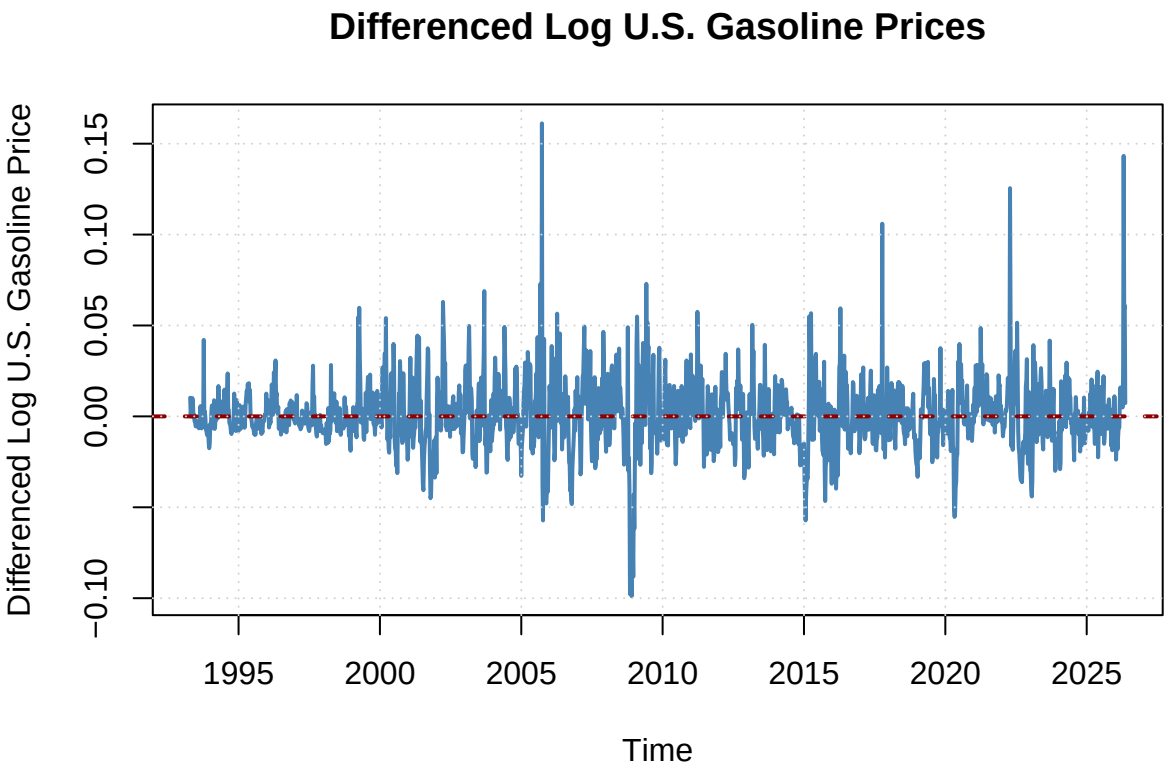

Figure 2: The logged and differenced time series for U.S. gasoline prices appears approximately stationary.

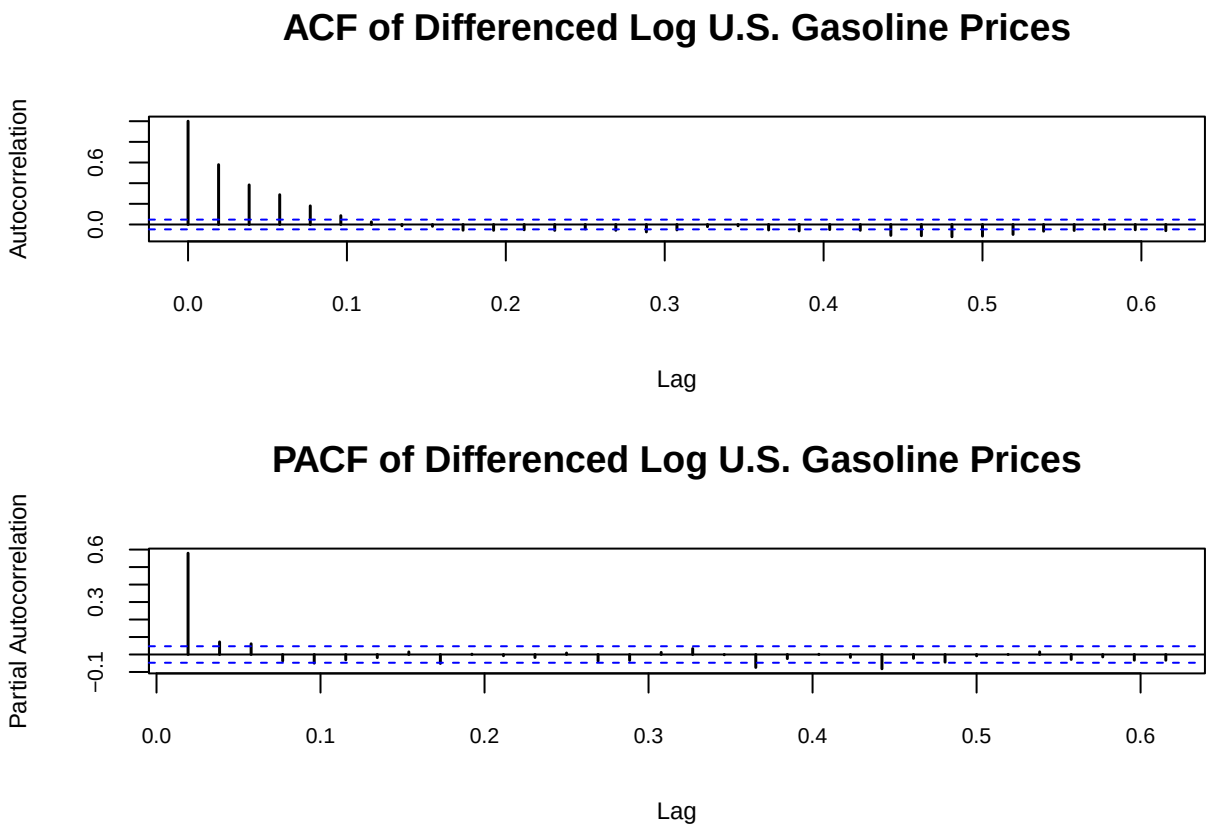

Figure 3: The ACF and PACF plots suggest a ARIMA(p,d,q) model would be a more appropriate fit for the transformed U.S. gasoline prices.

The fitted ARIMA(4,0,1) model for U.S. gasoline prices is

$$x_t = 0.0007 + 1.4864x_{t-1} - 0.4766x_{t-2} + 0.0610x_{t-3} - 0.1000x_{t-4} + w_t - 0.9597w_{t-1}.$$

The positive coefficient on x_{t-1} suggests that an increase in gasoline prices in one week tends to be followed by an increase the following week. The negative coefficient on x_{t-2} suggests that part of this effect reverses after two weeks. The coefficients on x_{t-3} and x_{t-4} are smaller, indicating weaker dependence on changes from three and four weeks earlier. Overall, the combination of positive and negative autoregressive coefficients suggests a short-run oscillating pattern in gasoline price changes.

The large negative moving average coefficient indicates that a positive shock in one week tends to be followed by a negative correction in the next week.

The diagnostic plots in Figure 4 do not suggest any major concerns with the fitted model. The observed and fitted values are reasonably well aligned, indicating that the model captures the main patterns in the differenced gasoline price series. The residual ACF and PACF show little remaining autocorrelation, with only a few isolated spikes outside the confidence bounds; given the large number of lags examined, a small number of such spikes is expected by chance.

The residuals over time contain some larger shocks, but these correspond to periods of unusually large gasoline price movements and likely reflect external events rather than model misspecification. The QQ plot indicates heavier tails than would be expected under a normal distribution. However, this behavior is common in financial and commodity price series and does not appear to indicate a serious problem with the model.

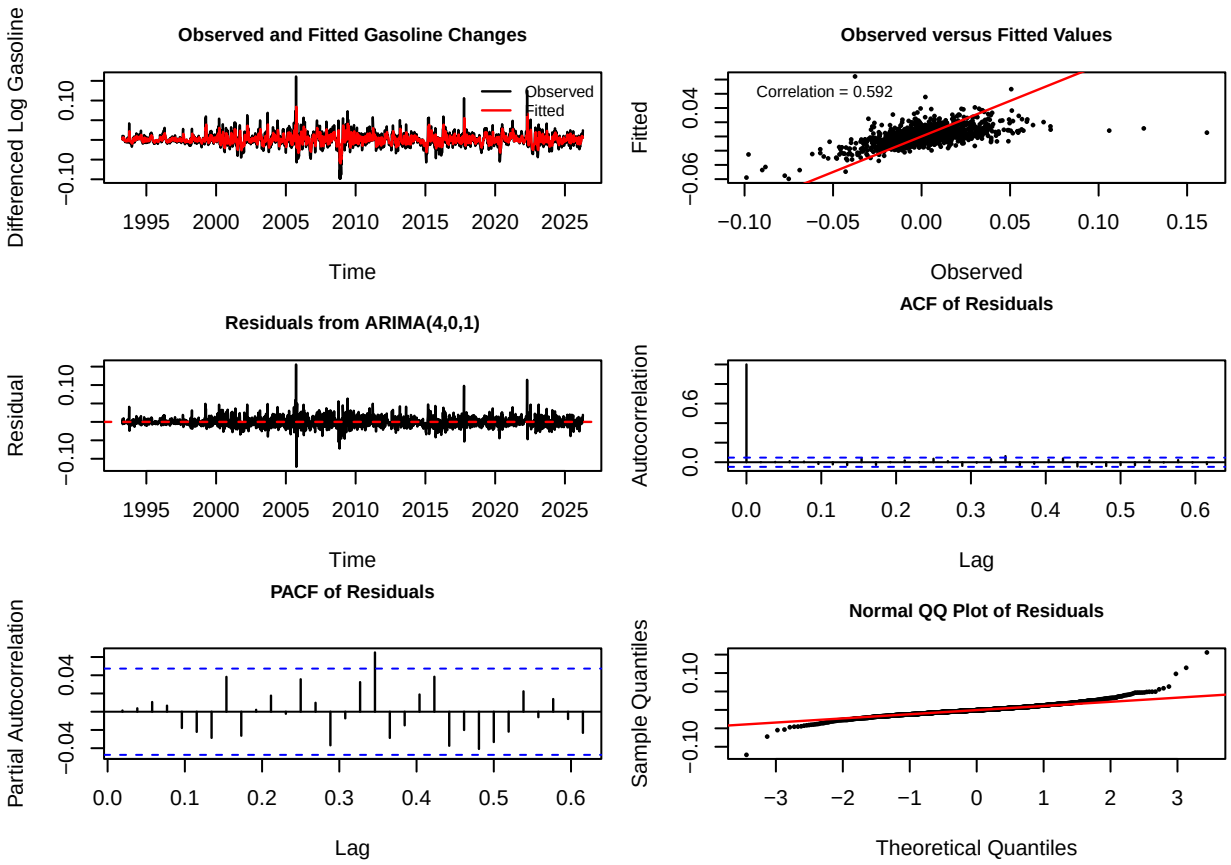

Figure 4: Diagnostics for ARIMA(4,0,1) of the transformed U.S. gasoline prices suggest no major concerns about the model fit.

The forecast in Figure 5 is particularly interesting because the holdout period occurs during a period of unusually high gasoline prices and substantial market shocks. As a result, the model does not fully capture the initial sharp increase in the logged and differenced gasoline price series. However, as gasoline prices begin to decline near the end of the four week forecast period, the forecasted values move much closer to the observed values. This suggests that the model is better able to capture the underlying dynamics of the series once the extreme shock begins to subside.

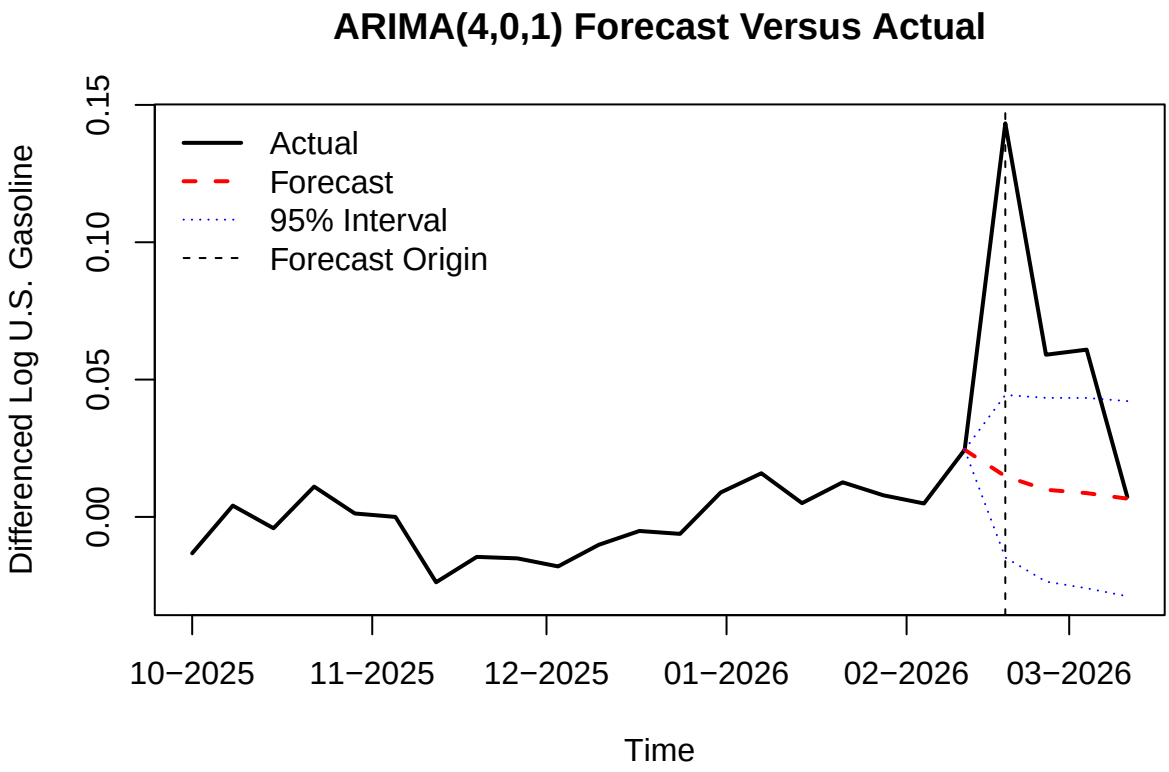

Figure 5: The ARIMA(4,0,1) model does not fully capture the unusually large shocks in gasoline prices, but its forecasts become much closer to the observed values once prices begin to decline.

3.2 Gasoline-Oil Prices VAR(2) Model

The Brent crude oil price series was log-transformed and differenced in the same manner as the gasoline price series and exhibited a similar approximate stationary structure. Based on Figure 6, the cross-correlation function peaks at lag 0, which means gasoline and Brent often move strongly within the same week. There are also positive correlations at negative lags, especially around -1 to -3, which is consistent with Brent changes showing up in gasoline prices over the following weeks.

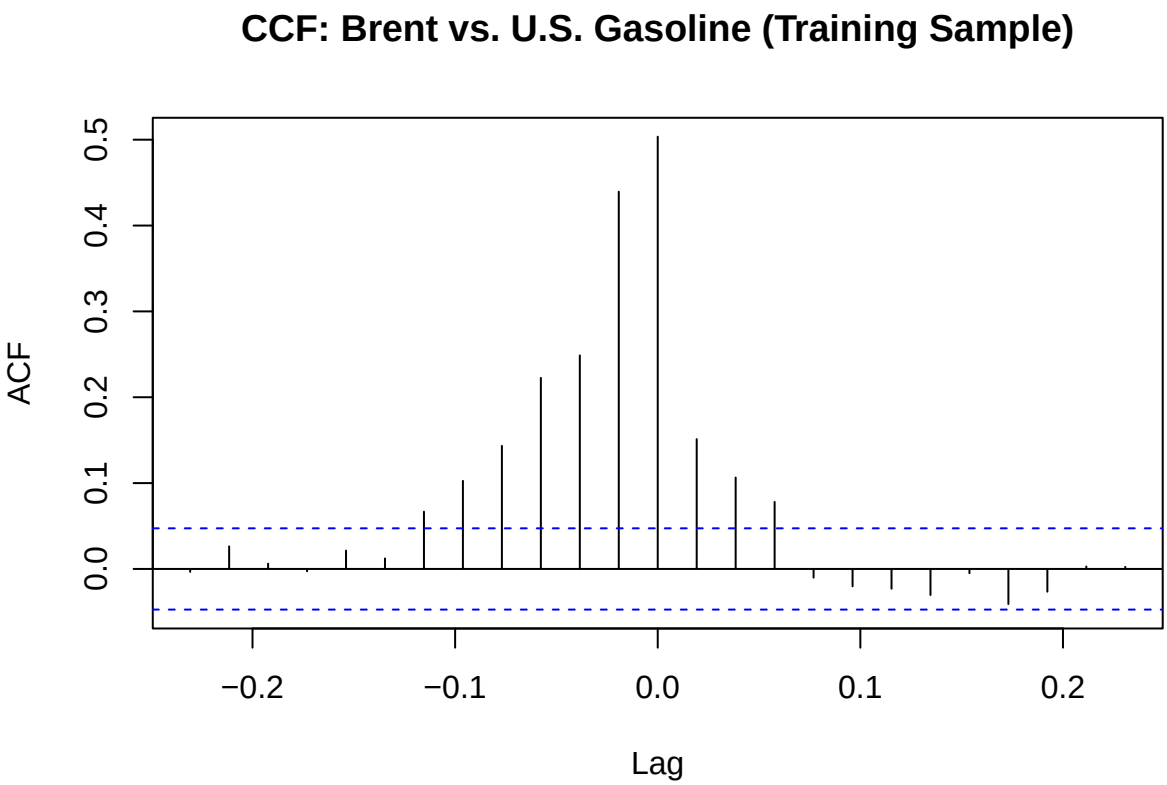

Figure 6: The CCF plot for transformed U.S. gasoline Brent crude oil prices suggests that they often move strongly within the same week and also Brent crude oil price changes reflected in gasoline prices over the following weeks.

From the available candidate models in Table 1, the lowest-order model, a VAR(2), provided sufficiently good diagnostic results and was therefore selected.

Table 1: Candidate model orders for VAR based on different information criteria. The simplest model was chosen for parsimony.

AIC(n)	HQ(n)	SC(n)	FPE(n)
5	3	2	5

The fitted VAR(2) model was

$$\text{Gasoline}_t = 0.4115\,\text{Gasoline}_{t-1} + 0.0946\,\text{Brent}_{t-1} + 0.1393\,\text{Gasoline}_{t-2} - 0.0242\,\text{Brent}_{t-2} + w_{1,t},$$

and

$$\text{Brent}_t = -0.0095\,\text{Gasoline}_{t-1} + 0.2547\,\text{Brent}_{t-1} + 0.2799\,\text{Gasoline}_{t-2} - 0.0926\,\text{Brent}_{t-2} + w_{2,t}.$$

In the gasoline equation, we can see clearly that gasoline prices are strongly driven by both their own past values and past Brent prices. The positive coefficients on gasoline at lags 1 and 2 suggest that increases in gasoline prices tend to persist for several weeks. The positive coefficient on Brent crude oil at lag 1 indicates that increases in crude oil prices are followed by increases in gasoline prices in the following week. The

negative coefficient on Brent at lag 2 suggests a partial correction after two weeks, so that some of the initial effect of a crude oil shock reverses over time.

Similarly, in the Brent equation, the positive coefficient on Brent at lag 1 suggests persistence in crude oil prices, while the negative coefficient on Brent at lag 2 suggests that part of this effect reverses after two weeks. The coefficient on gasoline at lag 1 is small and not statistically significant. However, gasoline at lag 2 appears as a statistically significant positive predictor of Brent prices. Gasoline prices do not logically drive crude oil prices, so the significance of lagged gasoline in the Brent equation should not be interpreted as a true causal effect. Instead, this likely reflects the fact that both series respond to common external factors, such as overall energy demand, supply disruptions, geopolitical events, or broader macroeconomic conditions. Because gasoline and crude oil prices move together over time, the model can identify statistically significant relationships even when they do not make economic sense as causal effects.

To further support this interpretation, the model fit appears stronger for the gasoline equation than for the Brent equation, as shown in Figure 7. The observed versus fitted plot for gasoline shows a higher correlation, indicating that the model explains a larger portion of the variation in gasoline prices. In contrast, the Brent equation has a weaker fit and the QQ plot for the Brent residuals shows greater deviation from normality.

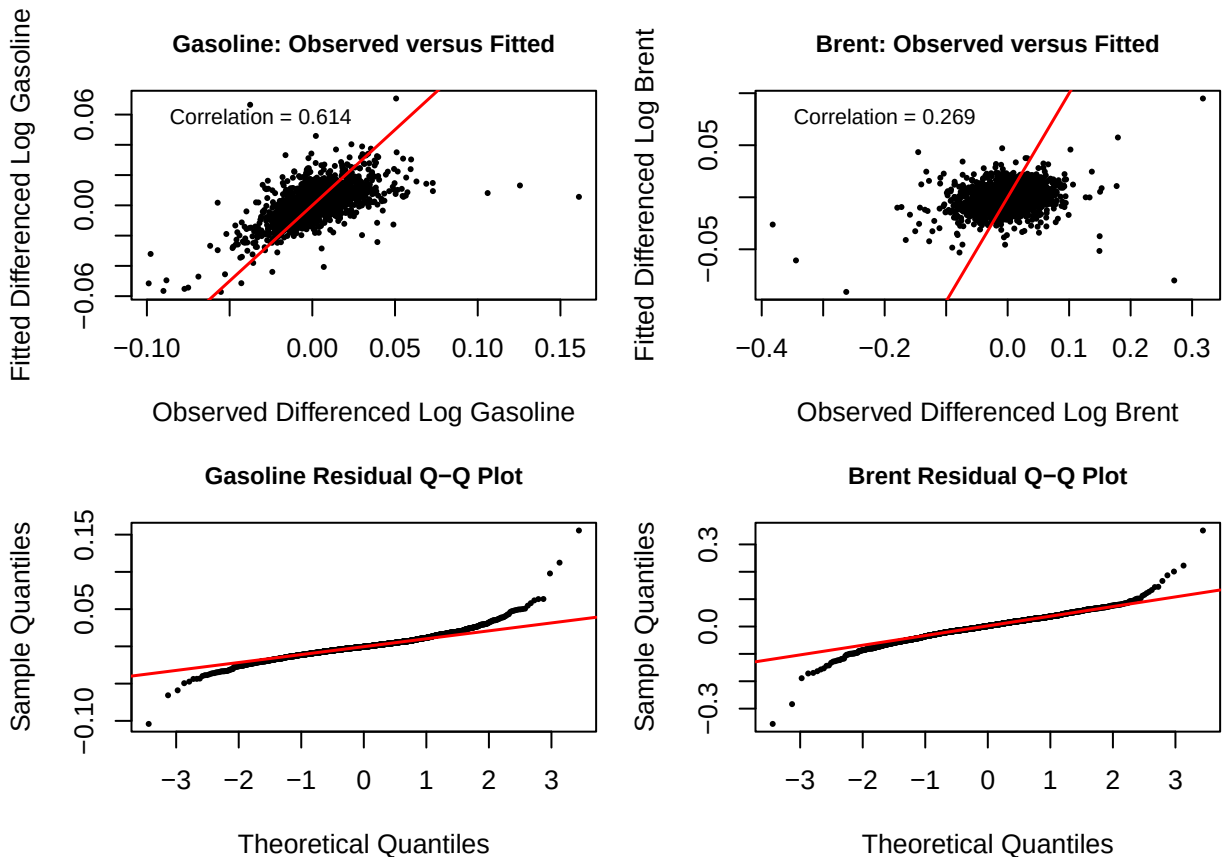

Figure 7: Diagnostics for VAR(2) of the transformed U.S. gasoline prices and Brent crude oils prices suggest a stronger model fit for gasoline prices.

Figure 8 shows the VAR(2) forecast is farther from the observed gasoline values than the ARIMA(4,0,1) forecast, even after gasoline prices begin to decline near the end of the forecast period. The VAR model predicts that the differenced gasoline series will quickly return toward zero.

This may occur because the VAR model incorporates Brent crude oil prices in addition to gasoline prices. During periods of large gasoline price shocks, the relationship between crude oil and gasoline may weaken.

Gasoline prices may be driven by additional factors such as refinery or transportation disruptions, local supply shortages or sudden changes in demand that are not captured by Brent crude oil prices alone.

In contrast, the ARIMA(4,0,1) model relies only on the recent history of gasoline prices. Because the recent gasoline observations remain elevated, the ARIMA model is better able to continue the high-price pattern into the forecast period and therefore remains closer to the observed values.

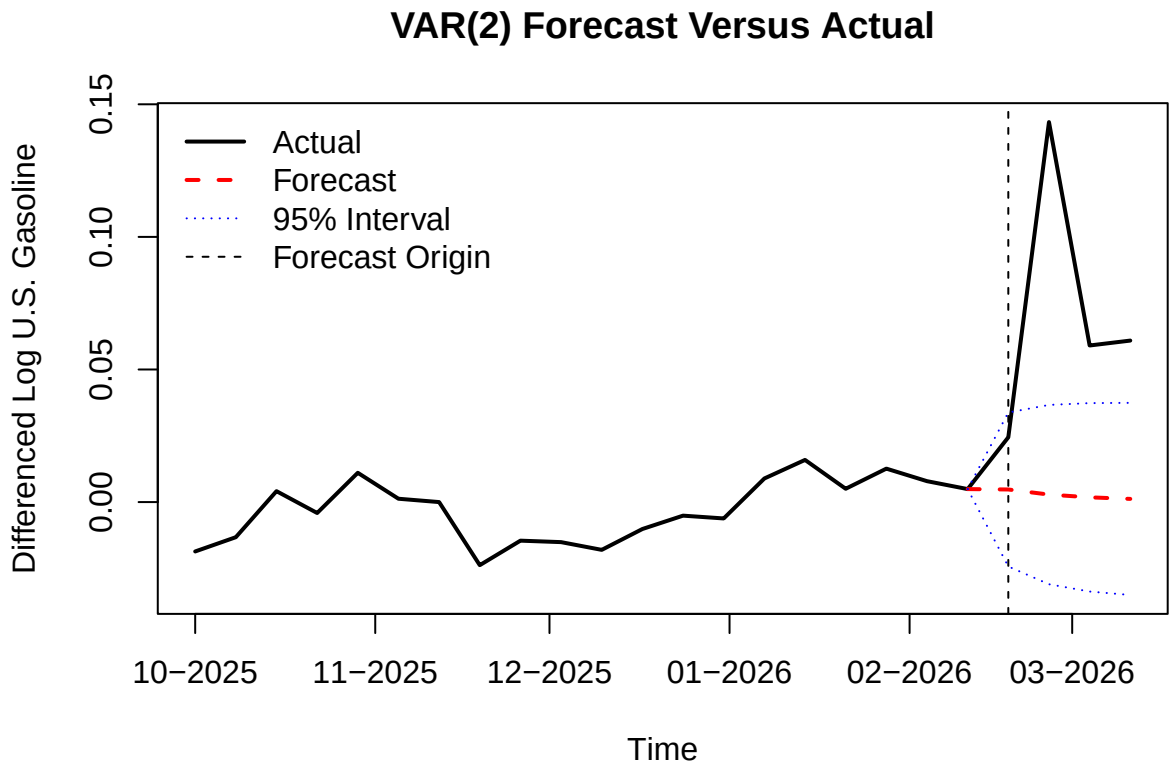

Figure 8: The VAR(2) model predicts further away than the ARIMA(4,0,1).

3.3 Regional Gasoline Prices VAR(1) Model

For the regional comparison, we identified a low order VAR(1) model using the same methods as for Brent crude oil versus U.S. gasoline prices. Table 2 shows the strongest regional lead-lag relationships. The CCF plots followed very similar structure to the one seen before with a mix of positive and negative lags. The fit of the VAR(1) model was validated by diagnostics which showed no reason for concern.

The regional VAR(1) model suggested that gasoline prices are highly interconnected across regions, with the strongest relationships centered around the Midwest, Gulf Coast, and West Coast. West Coast and Rocky Mountain prices were primarily driven by their own past values, indicating more localized dynamics.

Table 2: Of the strongest regional lead-lag relationships, the top autocorrelations are larger than the top cross-correlation coefficients from the VAR(1) model.

Type	Lagged Relationships	Coefficients
Auto-correlation	$West\ Coast_{t-1} \rightarrow West\ Coast_t$	0.567
Auto-correlation	$Midwest_{t-1} \rightarrow Midwest_t$	0.321
Cross-correlation	$Midwest_{t-1} \rightarrow Gulf\ Coast_t$	0.304
Cross-correlation	$East\ Coast_{t-1} \rightarrow Midwest_t$	-.303

Type	Lagged Relationships	Coefficients
Cross-correlation	$West\ Coast_{t-1} \rightarrow Rocky\ Mountains_t$	0.289

4 Discussion

4.1 ARIMA(4,0,1) Model

The forecast plot in Figure 5 shows the main limitation of this univariate model. The 4-week forecasts quickly revert toward zero and miss the sharp movement at the end of the sample. Even though the observed values remain inside the forecast interval, the point forecasts are too smooth to track the sudden change. This suggests that a model based only on past gasoline prices is not enough when market conditions shift quickly.

4.2 Diagnostics - VAR(2) Model

The largest residuals for U.S. gasoline prices in Figure 9 are due to external shocks that are outside the scope of this model. The first peak in October 2005 is due to Hurricane Katrina, which damaged oil refineries in the gulf coast, leading to spikes in gasoline prices. The shock at November 2008 is due to the financial crisis, which decreased gasoline demand and lead to lower prices. The residual at October 2017 corresponds to OPEC announcing supply cut for oil, leading to higher prices. In April 2022, sanctions on Russian oil lead to higher prices. Finally, during April 2026, oil prices have increased since the Straight of Hormuz is blocked. The residuals for Brent oil prices and regional gas prices have similarly timed residuals, corresponding to the same events.

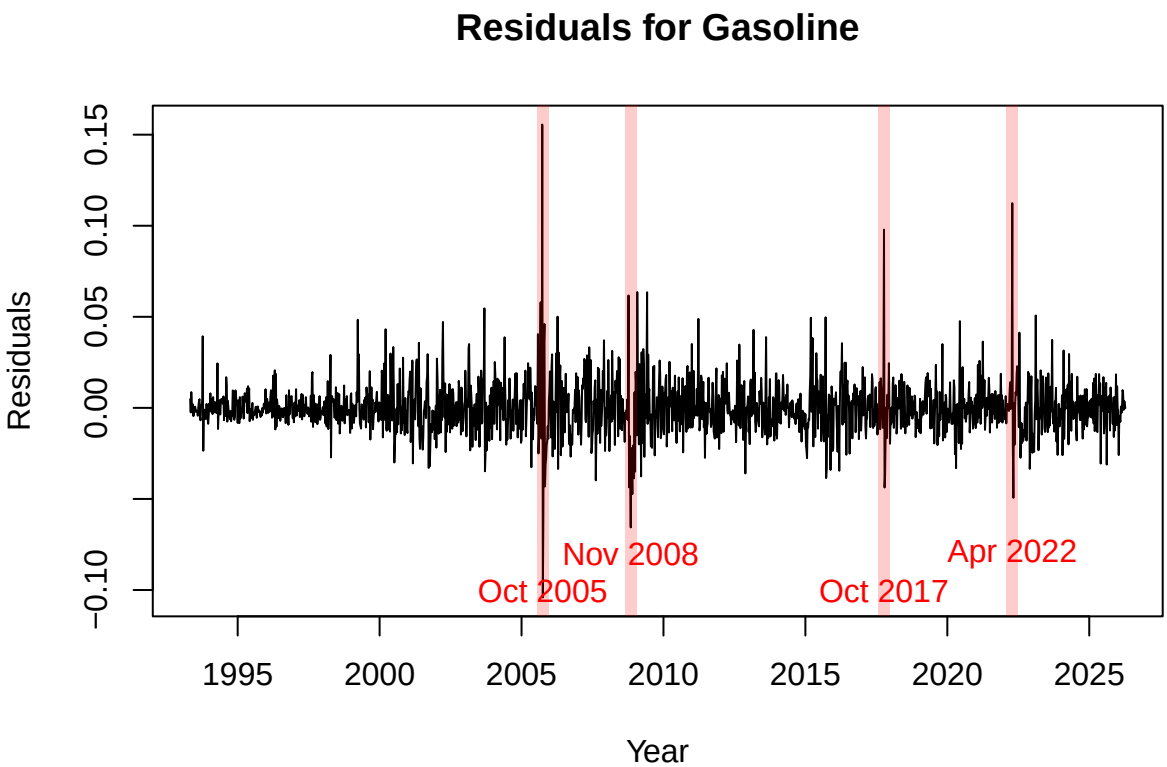

Figure 9: The largest residuals correspond to historical events that cannot be reliably modeled.

4.3 Diagnostics - VAR(1) Model

The residuals are a bit larger overall, especially during volatile periods, which suggests the model is not capturing everything. There is also more clustering in the residuals, meaning there are stretches where the model consistently under- or over-predicts. The same major events show up here as well, such as the spike from Hurrican Katrina in 2005 and the decline in demand and prices from the 2008 Financial Crisis, but with bigger errors, indicating that using only one lag is not enough to fully describe how prices evolve over time. There is also some remaining autocorrelation in the residuals, which again points to the model leaving some structure unexplained. Overall, the VAR(1) model works for the regional comparison.

4.4 Overall Interpretation

Putting everything together, a few key takeaways stand out. First, gasoline prices are closely tied to Brent crude oil, but the effect shows up with a short delay rather than immediately. Models that include oil prices capture this relationship better than those that do not. Second, gasoline prices across regions generally move together, which suggests a well-connected market. That said, regions like the West Coast and Rocky Mountains behave somewhat differently, likely due to local constraints like regulations and infrastructure. Third, all models struggle during periods of extreme volatility. The largest errors line up with major global events, showing that models based only on past prices miss important external factors. Overall, while the VAR(2) model takes into account oil prices and shows they are a significant predictor of gasoline prices, the ARIMA(4,0,1) model actually does a better job capturing the structure in the data when doing forecasts. Adding external variables would likely improve performance, especially during large shocks.

5 References

2026. *Yahoo! Finance*. Yahoo! https://finance.yahoo.com/.

Aroussi, Ran. 2026. "Ranaroussi/Yfinance: Download Market Data from Yahoo! Finance's API." *GitHub*. https://github.com/ranaroussi/yfinance.

U.S. Energy Information Administration. 2026. "Weekly U.S. Retail Gasoline and Diesel Prices." https://www.eia.gov/dnav/pet/pet_pri_gnd_dcus_nus_w.htm.

Appendix

```r
###########################
# STYLE EDITS: IGNORE THIS
###########################

# normally you'll want to include this with the libraries at the beginning of your document
knitr::opts_chunk$set(message = FALSE) # include this if you don't want markdown to knit messages
knitr::opts_chunk$set(warning = FALSE) # include this if you don't want markdown to knit warnings
knitr::opts_chunk$set(echo = FALSE) # set echo = FALSE to hide code from output

# DATA CLEANING

library(here)
library(readr)
library(readxl)
library(dplyr)
library(tidyr)
library(ggplot2)
library(GGally)
library(vars)
library(forecast)
library(lubridate)
library(tibble)

# File paths
gas_file_path <- here("data", "PET_PRI_GND_A_EPM0_PTE_DPGAL_W.xls")
brent_file_path <- here("data", "brent_crude_weekly_1993_present.csv")

# Load weekly U.S. gasoline prices
us_gas_df <- read_excel(gas_file_path, sheet = "Data 1", skip = 2) |>
  rename(
    Date = 1,
    US_All_Grades = 2
  ) |>
  mutate(
    Date = as.Date(Date),
    US_All_Grades = as.numeric(US_All_Grades)
  ) |>
  arrange(Date)

# Load weekly regional gasoline prices
region_gas_df <- read_excel(gas_file_path, sheet = "Data 2", skip = 2) |>
  rename(
    Date = 1,
    East_Coast = 2,
    New_England = 3,
    Central_Atlantic = 4,
    Lower_Atlantic = 5,
    Midwest = 6,
    Gulf_Coast = 7,
    Rocky_Mountain = 8,
    West_Coast = 9,
```

```r
    West_Coast_ex_CA = 10
  ) |>
  mutate(
    Date = as.Date(Date),
    across(-Date, as.numeric)
  ) |>
  arrange(Date)

# Keep the five regions used in the regional VAR
gas_df <- us_gas_df |>
  left_join(
    region_gas_df |>
      dplyr::select(Date, East_Coast, Midwest, Gulf_Coast, Rocky_Mountain, West_Coast),
    by = "Date"
  ) |>
  drop_na()

# Load weekly Brent crude oil prices
brent_df <- read_csv(brent_file_path, show_col_types = FALSE) |>
  rename(Brent = Brent_Price_USD) |>
  mutate(
    Date = as.Date(Date),
    Brent = as.numeric(Brent)
  ) |>
  arrange(Date)

# Helper to create weekly ts objects
make_weekly_ts <- function(x, start_date) {
  ts(
    x,
    start = c(
      as.numeric(format(start_date, "%Y")),
      as.numeric(format(start_date, "%U")) + 1
    ),
    frequency = 52
  )
}

gas_start_date <- min(gas_df$Date)

# Univariate gasoline series
us_gas_ts <- make_weekly_ts(gas_df$US_All_Grades, gas_start_date)
east_ts <- make_weekly_ts(gas_df$East_Coast, gas_start_date)
midwest_ts <- make_weekly_ts(gas_df$Midwest, gas_start_date)
gulf_ts <- make_weekly_ts(gas_df$Gulf_Coast, gas_start_date)
rocky_ts <- make_weekly_ts(gas_df$Rocky_Mountain, gas_start_date)
west_ts <- make_weekly_ts(gas_df$West_Coast, gas_start_date)

# Multivariate gasoline series for regional VAR analysis
region_var_ts <- gas_df |>
  dplyr::select(East_Coast, Midwest, Gulf_Coast, Rocky_Mountain, West_Coast) |>
  as.matrix() |>
  ts(
```

```r
    start = c(
      as.numeric(format(gas_start_date, "%Y")),
      as.numeric(format(gas_start_date, "%U")) + 1
    ),
    frequency = 52
  )

colnames(region_var_ts) <- c(
  "East_Coast", "Midwest", "Gulf_Coast", "Rocky_Mountain", "West_Coast"
)
# Align gasoline and Brent crude on common weekly dates
gas_brent_df <- gas_df |>
  inner_join(brent_df, by = "Date") |>
  drop_na()
gas_brent_start_date <- min(gas_brent_df$Date)

gas_ts_brent_overlap <- make_weekly_ts(gas_brent_df$US_All_Grades, gas_brent_start_date)
brent_ts <- make_weekly_ts(gas_brent_df$Brent, gas_brent_start_date)
# EXPLORATORY DATA ANALYSIS

par(mfrow = c(1,2))
# Standardize gasoline and Brent for visual comparison
gas_scaled_ts <- ts(
  as.numeric(scale(as.numeric(gas_ts_brent_overlap))),
  start = start(gas_ts_brent_overlap),
  frequency = frequency(gas_ts_brent_overlap)
)

brent_scaled_ts <- ts(
  as.numeric(scale(as.numeric(brent_ts))),
  start = start(brent_ts),
  frequency = frequency(brent_ts)
)

ts.plot(
  brent_scaled_ts, gas_scaled_ts,
  col = c("steelblue", "firebrick"), lty = 1,
  main = "Standardized Brent Crude Oil\nand U.S. Gasoline Prices",
  xlab = "Year",
  ylab = "Standardized Value"
)
legend(
  "topleft",
  legend = c("Brent crude oil", "U.S. gasoline"),
  col = c("steelblue", "firebrick"),
  lty = 1,
  bty = "n",
  cex = 0.5
)
# Regional gasoline prices
ts.plot(
  east_ts, midwest_ts, gulf_ts, rocky_ts, west_ts,
  col = 1:5, lty = 1,
```

```r
  main = "Regional Weekly Gasoline Prices",
  xlab = "Year",
  ylab = "Dollars per Gallon"
)
legend(
  "topleft",
  legend = c("East Coast", "Midwest", "Gulf Coast", "Rocky Mountain", "West Coast"),
  col = 1:5, lty = 1, bty = "n",
  cex = 0.5
)
d_us_gas <- diff(log(us_gas_ts))
gas_diff_dates <- us_gas_df$Date[-1]

# Differenced log gasoline prices plots
plot(
  d_us_gas,
  type = "l",
  col = "steelblue",
  lwd = 2,
  main = "Differenced Log U.S. Gasoline Prices",
  xlab = "Time",
  ylab = "Differenced Log U.S. Gasoline Price"
)

abline(h = 0, col = "darkred", lwd = 2, lty = 2)
grid(col = "lightgray", lty = "dotted")

# mtext(
#   "The series appears approximately stationary.",
#   side = 1,
#   line = 4,
#   cex = 0.9,
#   col = "gray30"
# )
# ACF and PACF

par(mfrow = c(2, 1), mar = c(4, 4, 4, 1))

acf(
  d_us_gas,
  main = "ACF of Differenced Log U.S. Gasoline Prices",
  xlab = "Lag",
  ylab = "Autocorrelation",
  lwd = 1.5,
  cex.main = 0.8,
  cex.lab = 0.75,
  cex.axis = 0.7
)

pacf(
  d_us_gas,
  main = "PACF of Differenced Log U.S. Gasoline Prices",
  xlab = "Lag",
```

```r
  ylab = "Partial Autocorrelation",
  lwd = 1.5,
  cex.main = 0.8,
  cex.lab = 0.75,
  cex.axis = 0.7
)

par(mfrow = c(1, 1))
# Hold out the last 4 weeks for a simple forecast check
forecast_horizon <- 4
n_gas <- length(d_us_gas)

train_d_us_gas <- window(d_us_gas, end = time(d_us_gas)[n_gas - forecast_horizon])
test_d_us_gas <- window(d_us_gas, start = time(d_us_gas)[n_gas - forecast_horizon + 1])
auto.arima(train_d_us_gas)
fit_arima_gas <- arima(train_d_us_gas, order = c(4, 0, 1))
# Fitted values and residuals for ARIMA(4,0,1)
fitted_gas_401 <- fitted(fit_arima_gas)

fit_gas_401 <- ts(
  as.numeric(fitted_gas_401),
  start = start(train_d_us_gas),
  frequency = frequency(train_d_us_gas)
)

obs_gas_401 <- window(
  train_d_us_gas,
  start = start(fit_gas_401),
  end = end(fit_gas_401)
)

gas_resid_401 <- residuals(fit_arima_gas)

par(mfrow = c(3, 2), mar = c(4, 4, 3, 1), cex.main = 0.85)

# 1. Observed vs fitted over time
plot(
  obs_gas_401,
  type = "l",
  col = "black",
  lwd = 1.2,
  main = "Observed and Fitted Gasoline Changes",
  xlab = "Time",
  ylab = "Differenced Log Gasoline"
)
lines(
  fit_gas_401,
  col = "red",
  lwd = 1.2
)
legend(
  "topright",
  legend = c("Observed", "Fitted"),
```

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  col = c("black", "red"),
  lwd = 1.2,
  bty = "n",
  cex = 0.75
)

# 2. Observed versus fitted scatterplot
plot(
  as.numeric(obs_gas_401),
  as.numeric(fit_gas_401),
  pch = 16,
  cex = 0.5,
  main = "Observed versus Fitted Values",
  xlab = "Observed",
  ylab = "Fitted"
)
abline(0, 1, col = "red", lwd = 1.2)
legend(
  "topleft",
  legend = paste(
    "Correlation =",
    round(cor(as.numeric(obs_gas_401), as.numeric(fit_gas_401)), 3)
  ),
  bty = "n",
  cex = 0.75
)

# 3. Residuals over time
plot(
  gas_resid_401,
  type = "l",
  col = "black",
  lwd = 1.2,
  main = "Residuals from ARIMA(4,0,1)",
  xlab = "Time",
  ylab = "Residual"
)
abline(h = 0, col = "red", lwd = 1.2, lty = 2)

# 4. Residual ACF
acf(
  gas_resid_401,
  main = "ACF of Residuals",
  xlab = "Lag",
  ylab = "Autocorrelation",
  lwd = 1.2
)

# 5. Residual PACF
pacf(
  gas_resid_401,
  main = "PACF of Residuals",
  xlab = "Lag",
```

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  ylab = "Partial Autocorrelation",
  lwd = 1.2
)

# 6. Residual QQ plot
qqnorm(
  gas_resid_401,
  main = "Normal QQ Plot of Residuals",
  xlab = "Theoretical Quantiles",
  ylab = "Sample Quantiles",
  pch = 16,
  cex = 0.5
)
qqline(
  gas_resid_401,
  col = "red",
  lwd = 1.2
)

par(mfrow = c(1, 1))

# 4-week forecast from the fitted ARIMA(4,0,1) model
fc_gas <- predict(fit_arima_gas, n.ahead = forecast_horizon)

# Forecast vectors
fc_mean <- as.numeric(fc_gas$pred)
fc_se <- as.numeric(fc_gas$se)
fc_lower <- fc_mean - 1.96 * fc_se
fc_upper <- fc_mean + 1.96 * fc_se

# Actual dates corresponding to the differenced gasoline series
train_dates <- gas_diff_dates[1:(n_gas - forecast_horizon)]
test_dates  <- gas_diff_dates[(n_gas - forecast_horizon + 1):n_gas]

# Show the last 20 weeks before the forecast origin plus the test period
weeks_of_history <- 20
tail_idx_start <- n_gas - forecast_horizon - weeks_of_history + 1

actual_tail_y <- d_us_gas[tail_idx_start:(n_gas - forecast_horizon)]
actual_tail_dates <- train_dates[(length(train_dates) - weeks_of_history + 1):length(train_dates)]

plot(
  x = actual_tail_dates,
  y = actual_tail_y,
  type = "l",
  lwd = 2,
  col = "black",
  main = "ARIMA(4,0,1) Forecast Versus Actual",
  xlab = "Time",
  ylab = "Differenced Log U.S. Gasoline",
  xlim = c(min(actual_tail_dates), max(test_dates)),
  ylim = range(
    c(actual_tail_y, test_d_us_gas, fc_mean, fc_lower, fc_upper),
```

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    na.rm = TRUE
  ),
  xaxt = "n"
)

tick_pos <- seq(from = min(actual_tail_dates), to = max(test_dates), by = "1 month")
axis(1, at = tick_pos, labels = format(tick_pos, "%m-%Y"))

# Actual held-out values
lines(
  x = c(tail(actual_tail_dates, 1), test_dates),
  y = c(tail(actual_tail_y, 1), test_d_us_gas),
  lwd = 2,
  col = "black"
)

# Forecast mean and 95% interval
lines(
  x = c(tail(actual_tail_dates, 1), test_dates),
  y = c(tail(actual_tail_y, 1), fc_mean),
  lwd = 2,
  lty = 2,
  col = "red"
)

lines(
  x = c(tail(actual_tail_dates, 1), test_dates),
  y = c(tail(actual_tail_y, 1), fc_lower),
  lty = 3,
  col = "blue"
)

lines(
  x = c(tail(actual_tail_dates, 1), test_dates),
  y = c(tail(actual_tail_y, 1), fc_upper),
  lty = 3,
  col = "blue"
)

# Forecast origin
abline(v = test_dates[1], lty = 2)

legend(
  "topleft",
  legend = c("Actual", "Forecast", "95% Interval", "Forecast Origin"),
  col = c("black", "red", "blue", "black"),
  lty = c(1, 2, 3, 2),
  lwd = c(2, 2, 1, 1),
  bty = "n"
)
# First differences of log prices
d_us_gas_overlap <- diff(log(gas_ts_brent_overlap))
d_brent <- diff(log(brent_ts))
```

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# Calendar dates corresponding to the differenced series
gas_brent_diff_dates <- gas_brent_df$Date[-1]
# Hold out the last 4 weeks for forecasting to match the ARIMA analysis
forecast_horizon <- 4
n_overlap <- length(d_us_gas_overlap)

train_d_us_gas_overlap <- window(
  d_us_gas_overlap,
  end = time(d_us_gas_overlap)[n_overlap - forecast_horizon]
)

test_d_us_gas_overlap <- window(
  d_us_gas_overlap,
  start = time(d_us_gas_overlap)[n_overlap - forecast_horizon + 1]
)

train_d_brent <- window(
  d_brent,
  end = time(d_brent)[n_overlap - forecast_horizon]
)

test_d_brent <- window(
  d_brent,
  start = time(d_brent)[n_overlap - forecast_horizon + 1]
)
# Differenced log oil prices plots
plot(
  d_brent,
  type = "l",
  col = "steelblue",
  lwd = 2,
  main = "Differenced Log Brent Crude Oil Prices",
  xlab = "Time",
  ylab = "Differenced Log Brent Crude Oil Price"
)

abline(h = 0, col = "darkred", lwd = 2, lty = 2)
grid(col = "lightgray", lty = "dotted")

mtext(
  "The series appears approximately stationary.",
  side = 1,
  line = 4,
  cex = 0.9,
  col = "gray30"
)
# Negative lag means Brent leads gasoline
# Use only the training sample when examining CCF
ccf(
  train_d_brent,
  train_d_us_gas_overlap,
  lag.max = 12,
  main = "CCF: Brent vs. U.S. Gasoline (Training Sample)"
```

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)
# Bivariate stationary training series for VAR analysis
oil_var_train_ts <- cbind(
  Gasoline = train_d_us_gas_overlap,
  Brent = train_d_brent
)
var_analysis <- VARselect(oil_var_train_ts, lag.max = 8, type = "none")
fit_var_oil <- VAR(oil_var_train_ts, p = 2, type = "none")
# Fitted values and residuals from the VAR(2) model
fitted_oil <- fitted(fit_var_oil)
resid_var <- residuals(fit_var_oil)

# Gasoline observed, fitted, and residuals
obs_gas_var <- window(
  oil_var_train_ts[, "Gasoline"],
  start = time(oil_var_train_ts)[3]
)

fit_gas_var <- ts(
  fitted_oil[, "Gasoline"],
  start = start(obs_gas_var),
  frequency = frequency(obs_gas_var)
)

resid_gas_var <- resid_var[, "Gasoline"]

# Brent observed, fitted, and residuals
obs_brent_var <- window(
  oil_var_train_ts[, "Brent"],
  start = time(oil_var_train_ts)[3]
)

fit_brent_var <- ts(
  fitted_oil[, "Brent"],
  start = start(obs_brent_var),
  frequency = frequency(obs_brent_var)
)

resid_brent_var <- resid_var[, "Brent"]

par(mfrow = c(2, 2), mar = c(4, 4, 3, 1), cex.main = 0.9)

# 1. Gasoline observed vs fitted
plot(
  as.numeric(obs_gas_var),
  as.numeric(fit_gas_var),
  pch = 16,
  cex = 0.5,
  main = "Gasoline: Observed versus Fitted",
  xlab = "Observed Differenced Log Gasoline",
  ylab = "Fitted Differenced Log Gasoline"
)
abline(0, 1, col = "red", lwd = 1.5)
```

```r
legend(
  "topleft",
  legend = paste(
    "Correlation =",
    round(cor(as.numeric(obs_gas_var), as.numeric(fit_gas_var)), 3)
  ),
  bty = "n",
  cex = 0.8
)

# 2. Brent observed vs fitted
plot(
  as.numeric(obs_brent_var),
  as.numeric(fit_brent_var),
  pch = 16,
  cex = 0.5,
  main = "Brent: Observed versus Fitted",
  xlab = "Observed Differenced Log Brent",
  ylab = "Fitted Differenced Log Brent"
)
abline(0, 1, col = "red", lwd = 1.5)
legend(
  "topleft",
  legend = paste(
    "Correlation =",
    round(cor(as.numeric(obs_brent_var), as.numeric(fit_brent_var)), 3)
  ),
  bty = "n",
  cex = 0.8
)

# 3. Gasoline QQ plot
qqnorm(
  resid_gas_var,
  main = "Gasoline Residual Q-Q Plot",
  xlab = "Theoretical Quantiles",
  ylab = "Sample Quantiles",
  pch = 16,
  cex = 0.5
)
qqline(resid_gas_var, col = "red", lwd = 1.5)

# 4. Brent QQ plot
qqnorm(
  resid_brent_var,
  main = "Brent Residual Q-Q Plot",
  xlab = "Theoretical Quantiles",
  ylab = "Sample Quantiles",
  pch = 16,
  cex = 0.5
)
qqline(resid_brent_var, col = "red", lwd = 1.5)
```

```r
par(mfrow = c(1, 1))

# 4-week ahead forecast
fc_var_oil <- predict(fit_var_oil, n.ahead = forecast_horizon)

# Extract forecast vectors (no ts())
fc_var_mean <- fc_var_oil$fcst$Gasoline[, "fcst"]
fc_var_lower <- fc_var_oil$fcst$Gasoline[, "lower"]
fc_var_upper <- fc_var_oil$fcst$Gasoline[, "upper"]

# Split actual dates
train_dates <- gas_brent_diff_dates[1:(n_overlap - forecast_horizon)]
test_dates  <- gas_brent_diff_dates[(n_overlap - forecast_horizon + 1):n_overlap]

# Last 20 weeks of training data
weeks_of_history <- 20
tail_idx_start <- n_overlap - forecast_horizon - weeks_of_history + 1

actual_tail_y <- d_us_gas_overlap[tail_idx_start:(n_overlap - forecast_horizon)]
actual_tail_dates <- train_dates[(length(train_dates) - weeks_of_history + 1):length(train_dates)]

# Plot using real dates
plot(
  x = actual_tail_dates,
  y = actual_tail_y,
  type = "l",
  lwd = 2,
  col = "black",
  main = "VAR(2) Forecast Versus Actual",
  xlab = "Time",
  ylab = "Differenced Log U.S. Gasoline",
  xlim = c(min(actual_tail_dates), max(test_dates)),
  ylim = range(
    c(actual_tail_y, test_d_us_gas_overlap, fc_var_mean, fc_var_lower, fc_var_upper),
    na.rm = TRUE
  ),
  xaxt = "n"
)

# Monthly axis (clean + readable)
tick_pos <- seq(from = min(actual_tail_dates), to = max(test_dates), by = "1 month")
axis(1, at = tick_pos, labels = format(tick_pos, "%m-%Y"))

# Actual held-out values
lines(
  x = c(tail(actual_tail_dates, 1), test_dates),
  y = c(tail(actual_tail_y, 1), test_d_us_gas_overlap),
  lwd = 2,
  col = "black"
)

# Forecast
lines(
```

```r
  x = c(tail(actual_tail_dates, 1), test_dates),
  y = c(tail(actual_tail_y, 1), fc_var_mean),
  lwd = 2,
  lty = 2,
  col = "red"
)

# Interval
lines(
  x = c(tail(actual_tail_dates, 1), test_dates),
  y = c(tail(actual_tail_y, 1), fc_var_lower),
  lty = 3,
  col = "blue"
)

lines(
  x = c(tail(actual_tail_dates, 1), test_dates),
  y = c(tail(actual_tail_y, 1), fc_var_upper),
  lty = 3,
  col = "blue"
)

# Forecast origin
abline(v = test_dates[1], lty = 2)

legend(
  "topleft",
  legend = c("Actual", "Forecast", "95% Interval", "Forecast Origin"),
  col = c("black", "red", "blue", "black"),
  lty = c(1, 2, 3, 2),
  lwd = c(2, 2, 1, 1),
  bty = "n"
)
d_regions <- diff(log(region_var_ts))
colnames(d_regions) <- colnames(region_var_ts)

region_diff_dates <- gas_df$Date[-1]

for (series_name in colnames(d_regions)) {
  plot(d_regions[, series_name],
       main = paste("Differenced Log", gsub("_", " ", series_name)),
       ylab = "Log difference",
       xlab = "Time")
  abline(h = 0, lty = 2, col = "gray50")

  par(mfrow = c(1, 2))
  acf(d_regions[, series_name],
      lag.max = 20,
      main = paste("ACF:", gsub("_", " ", series_name)))
  pacf(d_regions[, series_name],
       lag.max = 20,
       main = paste("PACF:", gsub("_", " ", series_name)))
  par(mfrow = c(1, 1))
```

```r
}

# Pairwise CCFs for all 10 regional pairs
region_pairs <- combn(colnames(d_regions), 2, simplify = FALSE)

# Show Midwest pairs first because they are especially informative in the plots
region_pairs <- c(
  Filter(function(x) "Midwest" %in% x, region_pairs),
  Filter(function(x) !("Midwest" %in% x), region_pairs)
)

region_pairs <- lapply(region_pairs, function(pair) {
  if ("Midwest" %in% pair) {
    c("Midwest", setdiff(pair, "Midwest"))
  } else {
    pair
  }
})

par(
  mfrow = c(2, 2),          # or another layout
  mar = c(3, 3, 2, 1),      # smaller margins
  oma = c(0, 0, 1, 0)
)

for (pair in region_pairs) {
  ccf(
    d_regions[, pair[1]],
    d_regions[, pair[2]],
    lag.max = 12,
    main = paste(
      gsub("_", " ", pair[1]),
      "vs.",
      gsub("_", " ", pair[2])
    ),
    cex.main = 0.9,
    cex.axis = 0.8,
    cex.lab = 0.8
  )
}

par(mfrow = c(1, 1))

region_model_ts <- d_regions

# Candidate lag lengths
VARselect(region_model_ts, lag.max = 8, type = "const")

# Fit the regional VAR(1)
fit_var_regions <- VAR(region_model_ts, p = 1, type = "const")
summary(fit_var_regions)

# Fitted values and residuals
```

```r
fitted_regions <- fitted(fit_var_regions)
resid_regions  <- resid(fit_var_regions)
p_used <- fit_var_regions$p

# Observed vs. fitted plots for each region
for (series_name in colnames(region_model_ts)) {
  obs_ts <- window(region_model_ts[, series_name], start = time(region_model_ts)[p_used + 1])
  fit_ts <- ts(fitted_regions[, series_name],
               start = start(obs_ts),
               frequency = frequency(obs_ts))

  par(mfrow = c(1, 2))

  plot(obs_ts, type = "l", col = "black",
       main = paste("Observed vs. Fitted:", gsub("_", " ", series_name)),
       ylab = "",
       xlab = "Time")
  lines(fit_ts, col = "red")

  plot(obs_ts, fit_ts,
       pch = 16,
       xlab = "Observed",
       ylab = "Fitted",
       main = gsub("_", " ", series_name))
  abline(0, 1, col = "red", lwd = 2)
  legend("topleft",
         legend = paste("Correlation:", round(cor(obs_ts, fit_ts), 3)),
         bty = "n")

  par(mfrow = c(1, 1))
}

# Residual series and ACFs
for (series_name in colnames(resid_regions)) {
  par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))

  plot(ts(resid_regions[, series_name],
          start = time(region_model_ts)[p_used + 1],
          frequency = frequency(region_model_ts)),
       type = "l",
       main = paste("Residuals:", gsub("_", " ", series_name)),
       ylab = "",
       xlab = "Time",
       ylim = c(-0.2, 0.2))
  abline(h = 0, lty = 2, col = "gray")

  acf(resid_regions[, series_name],
      main = paste("ACF Residuals:", gsub("_", " ", series_name)))

  par(mfrow = c(1, 1))
}
```

```r
library(tibble)

lead_lag_summary <- tibble(
  Type = c(
    "Autocorrelation",
    "Autocorrelation",
    "Cross-Correlation",
    "Cross-Correlation",
    "Cross-Correlation"
  ),
  `Lagged Relationship` = c(
    "West Coast[t-1] → West Coast[t]",
    "Midwest[t-1] → Midwest[t]",
    "Midwest[t-1] → Gulf Coast[t]",
    "East Coast[t-1] → Midwest[t]",
    "West Coast[t-1] → Rocky Mountain[t]"
  ),
  Coefficient = c(
    0.567,
    0.321,
    0.304,
    -0.303,
    0.289
  )
)
plot_residuals_with_events <- function(res_vec, dates_vec, series_name,
                                       threshold_upper, threshold_lower,
                                       label_y_positions) {

  # Build dataframe once
  df <- data.frame(time = dates_vec,
                   value = res_vec)

  # Filter event dates
  event_dates <- df |>
    filter(value > threshold_upper | value < threshold_lower) |>
    group_by(floor(time)) |>
    summarize(dates = mean(time), .groups = "drop") |>
    mutate(dates = as.Date(date_decimal(dates))) |>
    pull(dates)

  # Convert to decimal time
  event_times <- as.numeric(format(event_dates, "%Y")) +
    (as.numeric(format(event_dates, "%j")) - 1) / 365

  # Plot
  ts.plot(ts(res_vec,
             start = dates_vec[1],
             frequency = 52),
          lty = 1,
          main = paste("Residuals for", series_name),
          xlab = "Year", ylab = "Residuals")
```

```r
  usr <- par("usr")

  # Draw rectangles (vectorized)
  rect(xleft  = event_times - 0.2,
       xright = event_times + 0.2,
       ybottom = usr[3],
       ytop    = usr[4],
       col = rgb(1, 0, 0, alpha = 0.2),
       border = NA)

  # Labels
  y_positions <- rep(label_y_positions, length.out = length(event_times))

  text(x = event_times,
       y = y_positions,
       labels = format(event_dates, "%b %Y"),
       col = "red",
       cex = 1)
}

# Gasoline
plot_residuals_with_events(
  res_vec = resid_var[,1],
  dates_vec = time(oil_var_train_ts)[-c(1,2)],
  series_name = "Gasoline",
  threshold_upper = 0.08,
  threshold_lower = -0.05,
  label_y_positions = c(-0.1, -0.08)
)
```